ESA RN12 Environment & Society, Summer School for Early Career Researchers “Methodological Perspectives for Social Research on Energy and Environmental Issues”, 22-23 September, 2025

Venue: Faculty of Social Sciences, Arts and Humanities, Kaunas University of Technology

Opening Your Research Data in FAIR Way

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Topics

1. Why open data?
2. FAIR data principles for opening data
3. Data repositories for FAIR data

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https://lida.dataverse.lt

Why open data?

- Empirical science/research generates data
- Open data (science) – strategic priority
  - Not only publications should be open, but also data (software and other research outputs)
  - Goods created using public money (money from public funding agencies) should be public
- Data management plans (almost every public funding institution requires them)
  - How the data generated implementing projects are managed, where they are put and curated? Would they be opened (if possible)?

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Why open data?

- Basic principle of opening data
  - As Open as Possible, as Closed as Necessary
  - There are legitimate reasons for restricting openness of data
    - Short term – until a research group (researcher) finalizes publications and other outputs
    - Long term – due to confidentiality of the research subjects

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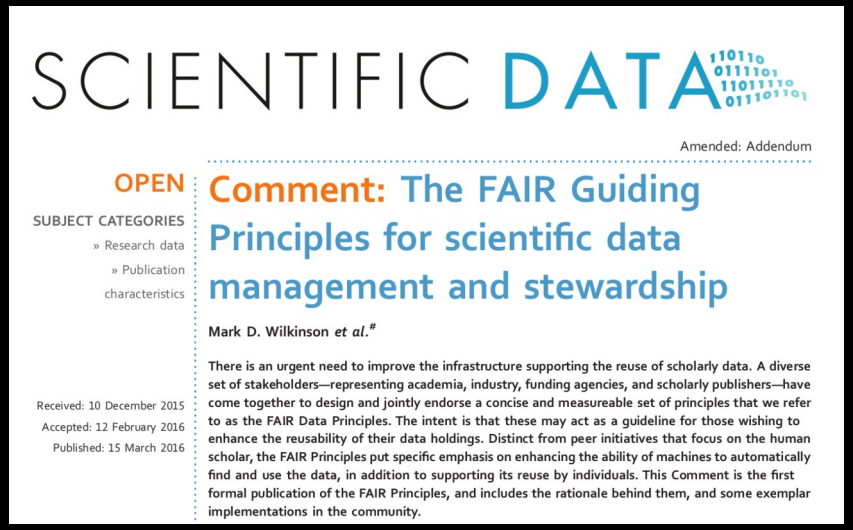
Why open data?


SCIENTIFIC DATA

Amended: Addendum

OPEN

SUBJECT CATEGORIES
» Research data
» Publication characteristics

Received: 10 December 2015
Accepted: 12 February 2016
Published: 15 March 2016

Comment: The FAIR Guiding Principles for scientific data management and stewardship

Mark D. Wilkinson *et al.*[#]

There is an urgent need to improve the infrastructure supporting the reuse of scholarly data. A diverse set of stakeholders—representing academia, industry, funding agencies, and scholarly publishers—have come together to design and jointly endorse a concise and measureable set of principles that we refer to as the FAIR Data Principles. The intent is that these may act as a guideline for those wishing to enhance the reusability of their data holdings. Distinct from peer initiatives that focus on the human scholar, the FAIR Principles put specific emphasis on enhancing the ability of machines to automatically find and use the data, in addition to supporting its reuse by individuals. This Comment is the first formal publication of the FAIR Principles, and includes the rationale behind them, and some exemplar implementations in the community.

- Opening data is simple (put somewhere on the web)?
  - Quality of opening data? (Open Access vs. Open/FAIR data)
  - In 2014 workshop in Leiden (the Netherlands) Jointly Designing a Data FAIRPORT
    - The meeting concluded with a draft formulation of a set of foundational principles that were subsequently elaborated in greater detail – namely, that all research objects should be Findable, Accessible, Interoperable and Reusable (FAIR) both for machines and for people. These are now referred to as the FAIR Guiding Principles (Wilkinson et al. 2016).

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FAIR data principles for opening data

- One more standard among many other, or something more “serious”?
  - Seems to be a long-term regulation
- What does it mean?
  - How to implement it (for researcher, science administrators etc.)
  - How to measure FAIRness?
- Research data will not become nor stay FAIR by magic. We need skilled people, transparent processes, interoperable technologies and collaboration to build, operate and maintain research data infrastructures.
  - Mari Kleemola, Finnish Social Science Data Archive/CoreTrustSeal Board, Secretary https://tietoarkistoblogi.blogspot.com/2018/11/being-trustworthy-and-fair.html

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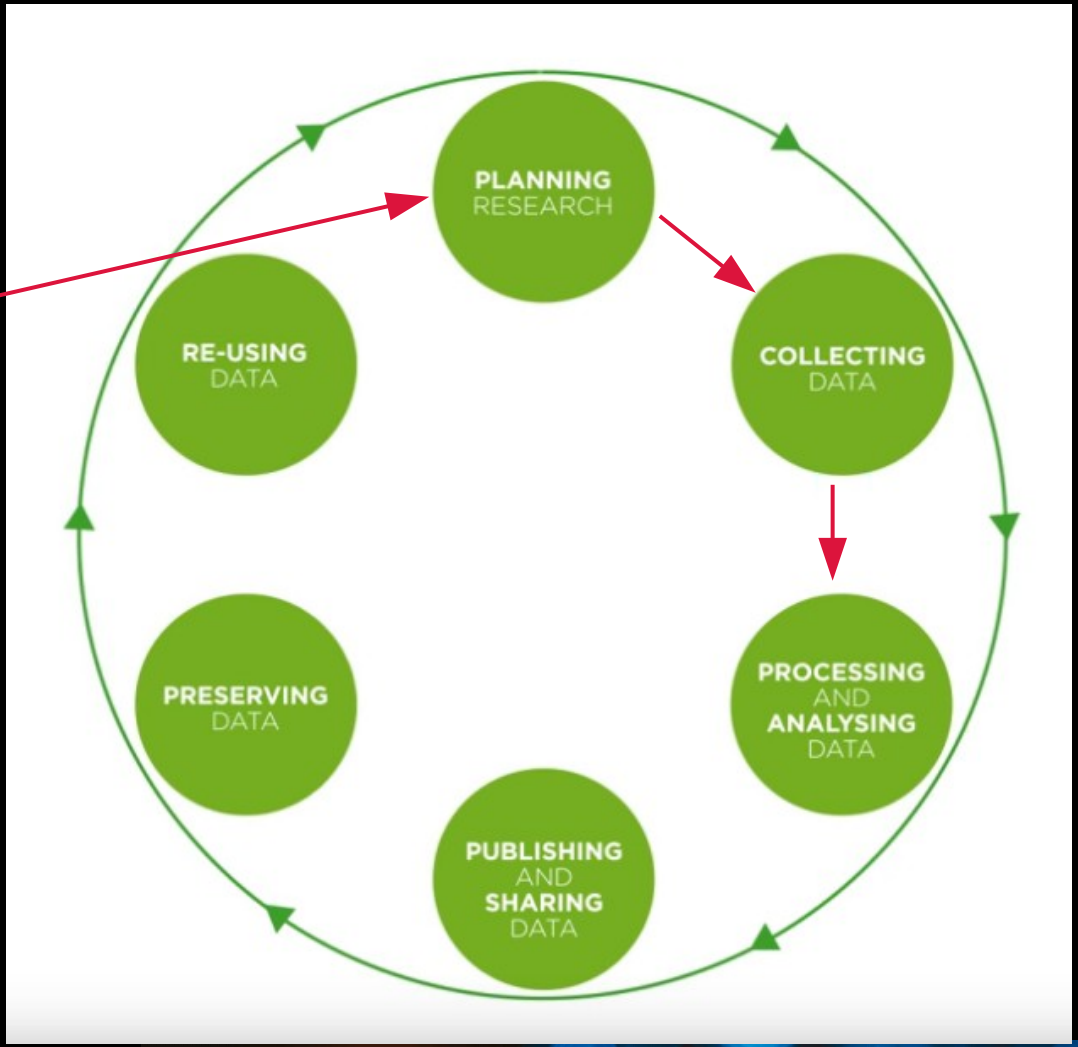
FAIR data principles for opening data

- Data lifecycle
  - Usually, researcher begins from the planing/design



UK Data Archive: https://ukdataservice.ac.uk/learning-hub/research-data-management

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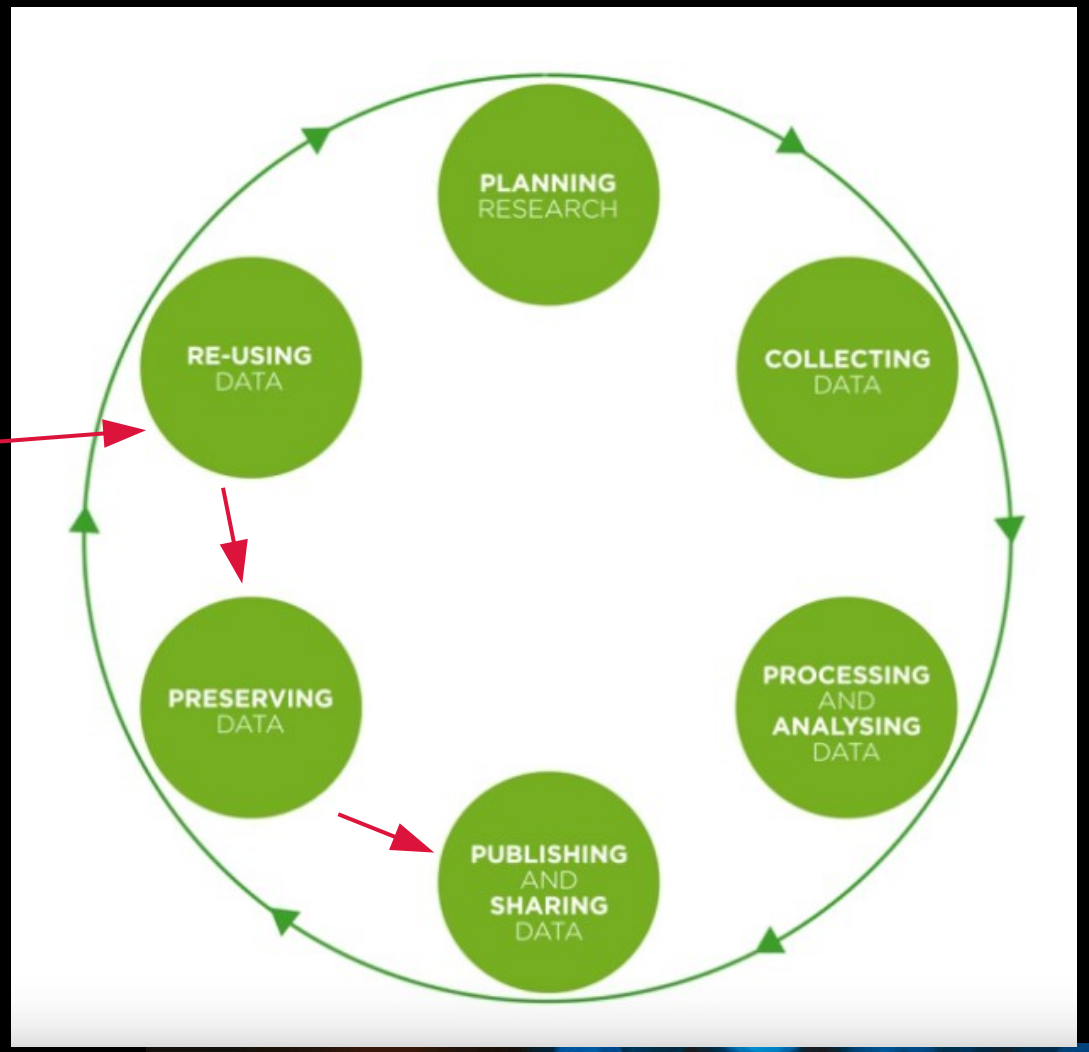
FAIR data principles for opening data

- Data lifecycle
  - Usually, researcher begins from the planing/design
  - However, the process of opening data starts at the last stage
    - Re-using data
    - You have to know your data, in order to use them properly



UK Data Archive: https://ukdataservice.ac.uk/learning-hub/research-data-management

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FAIR data principles for opening data

- Trustworthiness (trust in the used data) is most important
  - A person re-using the data needs to know
    - Where these data come from?
    - How they were generated?
    - How they were processed?
    - How they can be re-used?
  - A person who produced the data wants to be sure that
    - The generated data will be used and interpreted properly

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FAIR data principles for opening data

- FAIR data principles provide guidelines, how to make digital objects (for example, data sets) findable, accessible, interoperable and reusable during the whole data lifecycle
  - A FAIRy tale
    - https://forskningsdata.dk/fairytale
      - https://doi.org/10.5281/zenodo.2248199
    - www.snf.ch/SiteCollectionDocuments/FAIR_principles_translation_SNSF_logo.pdf

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FAIR data principles for opening data

www.go-fair.org/fair-principles, https://force11.org/info/the-fair-data-principles

- Findable
  - Most important – to find the data, so metadata and data should be easy to find for both humans and computers
  - Machine-readable metadata are essential for automatic discovery of datasets and services
    - F1. (Meta)data are assigned a globally unique and persistent identifier (PID → DOI, Handle, DataCite, B2HANDLE etc.)
    - F2. Data are described with rich metadata (as defined by R1)
    - F3. Metadata clearly and explicitly include the identifier of the data it describes
    - F4. (Meta)data are registered or indexed in a searchable resource (OpenAIRE, Google Dataset Search etc.)

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FAIR data principles for opening data

www.go-fair.org/fair-principles, https://force11.org/info/the-fair-data-principles

- Accessible
  - Once users find the required data, they need to know how they can be accessed, possibly including authentication and authorisation
  - A1. (Meta)data are retrievable by their identifier using a standardised communications protocol
    - A1.1. The protocol is open, free, and universally implementable (HTTP, FTP, phone etc.)
    - A1.2. The protocol allows for an authentication and authorisation procedure, where necessary (FAIR ≠ Open)
  - A2. Metadata are accessible, even when the data are no longer available (data should not (always) be open, just metadata)

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FAIR data principles for opening data

www.go-fair.org/fair-principles, https://force11.org/info/the-fair-data-principles

- Interoperable
  - The data usually need to be integrated with other data
  - In addition, the data need to interoperate with applications or workflows for analysis, storage, and processing
    - I1. (Meta)data use a formal, accessible, shared, and broadly applicable language for knowledge representation (XML, RDF)
    - I2. (Meta)data use vocabularies that follow FAIR principles (eg., https://vocabularies.cessda.eu)
    - I3. (Meta)data include qualified references to other (meta)data

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FAIR data principles for opening data

www.go-fair.org/fair-principles, https://force11.org/info/the-fair-data-principles

- Reusable
  - The ultimate goal of FAIR is to optimise the reuse of data
  - To achieve this, metadata and data should be well-described so that they can be replicated and/or combined in different settings
    - R1. (Meta)data are richly described with a plurality of accurate and relevant attributes
      - R1.1. (Meta)data are released with a clear and accessible data usage license (https://creativecommons.org/licenses)
      - R1.2. (Meta)data are associated with detailed provenance (how, why and by whom the data have been created and processed)
      - R1.3. (Meta)data meet domain-relevant community standards (https://rd-alliance.github.io/metadata-directory/standards, https://www.dcc.ac.uk/guidance/standards/metadata, https://fairsharing.org, https://dans-labs.github.io/formats)

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FAIR data principles for opening data

- How to measure FAIRness of the data? (https://github.com/FAIRMetrics/Metrics)
  - General Checklist by the EUDAT (https://doi.org/10.5281/zenodo.1065990)
  - FAIR Data Self-Assessment Tool by the ARDC (https://ardc.edu.au/resource/fair-data-self-assessment-tool)
  - Self-Assessment Tool to Improve the FAIRness of Your Dataset (SATIFYD) by the DANS (https://dans.knaw.nl/en/satifyd)

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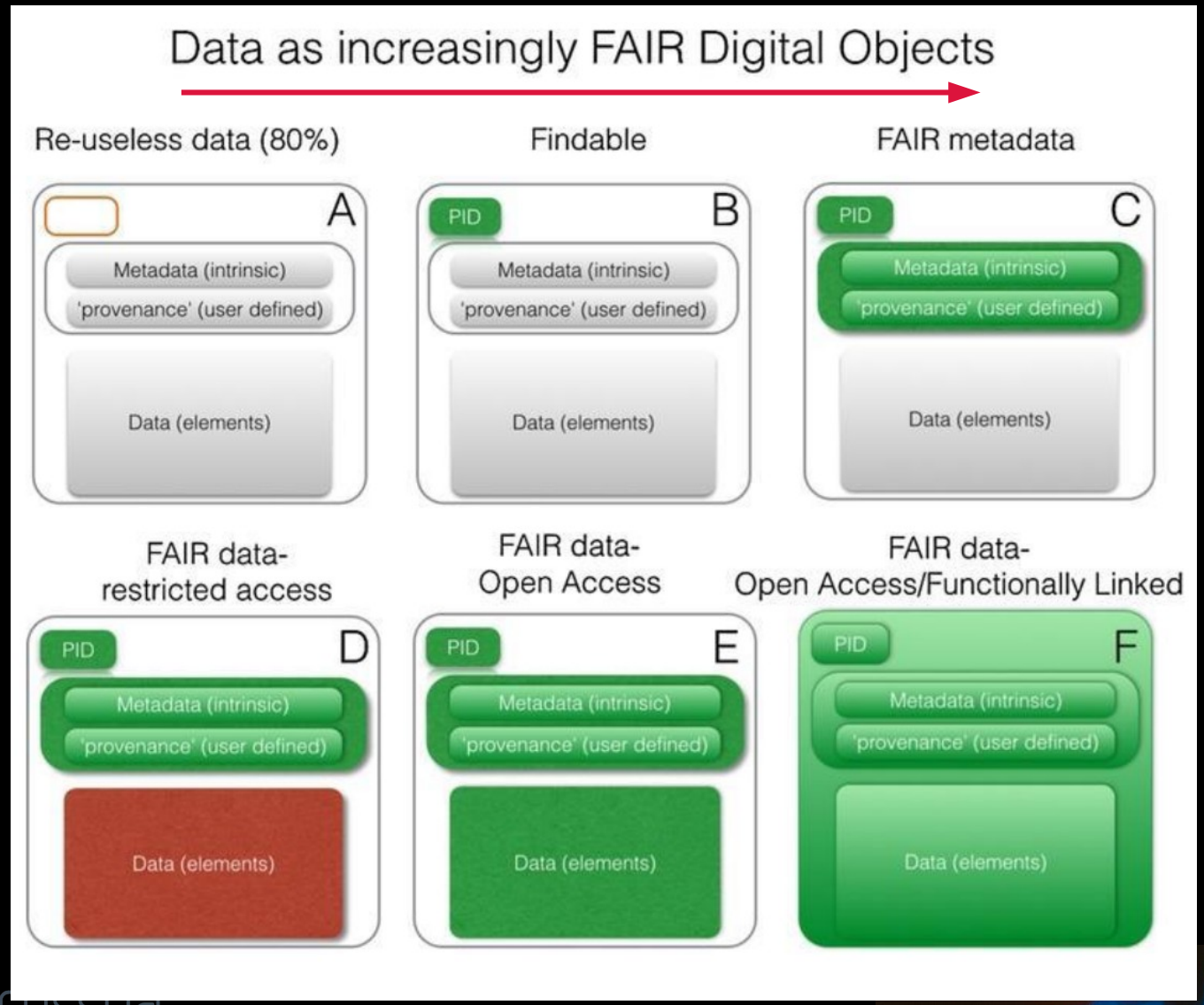
FAIR data principles for opening data



Mons et al., 2017

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Data repositories for FAIR data

- To make opened data compliant with FAIR principles is essentially impossible for an individual researcher
  - For this purpose multitude of data repositories were developed – virtual infrastructures, where data (and other digital objects) managed, stored, processed and published (curated)
    - Institutional (institution or department)
    - Domain specific (social science)
    - Generalist (zenodo.org)
  - All of them set certain requirements
    - Data re-use
    - File formats and data structure
    - Metadata standards

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Data repositories for FAIR data

- Why users (researchers) should worry about FAIR data, data repositories, and data repositories compliant with FAIR principles?
  - Advantages of data repositories
    - Researchers may worry less about proper opening of their data
    - Openness strengthens scientific integrity and trust in scientific products
    - Deposited data become
      - Accessible
      - Understandable
      - Re-usable
    - Data repositories make your data FAIR and curate them according to FAIR principles

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Data repositories for FAIR data

- How data become FAIR in repositories?
  - A few examples
    - PID is issued on publication
      - Long-term findability and proper citation
    - Findability via public search catalogues
      - Effective data findability is an essential premise of sharing data
    - Proper licensing of data and metadata
      - Clear terms and conditions of data (re)use
    - (Meta)data standards implementation and development
      - (Meta)data interoperability

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Data repositories for FAIR data

- How data is stored in repositories according to FAIR standards?
  - Data repositories ensure long-term curation of FAIR digital objects, ensuring that data remain accessible and FAIR-compliant over time
    - Advise and support data depositors (e.g. (meta)data standards, restricted access)
    - Advise and support data users (e.g. citation, how to use data)

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Data repositories for FAIR data

- How to find a suitable data repository?
  - Find a certified data repository
  - Use domain specific repository
    - Geriausiai išmano jums tinkamus (meta)duomenų standartus
  - Use institutional repository (if long-term preservation is available)
    - Sometimes more efective and convenient
  - Generalist repository – only is options above are not available
    - (Meta)data sandards are very general (suitable for all data types)
  - Searching for your repository
    - www.re3data.org (most comprehensive)
    - https://repositoryfinder.datacite.org

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Data repositories for FAIR data

- Trustworthy repositories usually are certified
  - Provides services that ensure compatibility of data sets with the FAIR principles
  - Assessed regularly according to established standards
  - There are several certification methods that can be used to assess the trustworthiness of a data repository (ISO, DIN)
    - The most commen certification system: CoreTrustSeal www.coretrustseal.org
  - Infrastructure is also important
    - www.dataverse.org platform is among the most-compliant with FAIR principles

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Data repositories for FAIR data

- Trustworthiness is the most important aspect when sharing and re-using data
- Measurement of FAIRness can be done at multiple levels and should include infrastructure (repository)
- FAIR-compliant data repositories ensure long-term data availability, understandability, and reusability
- FAIR data “live” in trusted data repositories
  - Certified repositories curate FAIR data according to FAIR principles
  - CoreTrustSeal requirements are compatible with FAIR data principles

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Data repositories for FAIR data

CORETRUSTSEAL— FAIR ALIGNMENT

F	R13	• Offer persistent identifiers **[F1 and F3]** • Recommended data citations **[F1]** • Searchable metadata catalogue to appropriate standards **[F2, F3]** • Search facilities, inclusion in disciplinary or generic registries of resources **[F4]**
A	R4 R10 R13 R15 R16	• Facilitate machine harvesting of the metadata **[A1]** • Uses international and/or community standards **[A1.1]** • Searchable metadata catalogue to appropriate standards **[A1 and A1.1]** • Technical infrastructure: protection of facility, data, products, services, users **[A1.2]** • Data managed in compliance with discipline and ethical norms **[A1.2]** • Responsibility for long-term preservation **[A2]**
I	R14 R11	• Metadata required when the data are provided **[I1]** • Formats used by the Designated Community **[I1]** • Measures and plans for the possible evolution and migration of formats **[I2]** • Ensure understandability of the data **[I2]** • Ability to comment on, and/or rate data and metadata **[I3]** • Provide citations to related works or links to citation indices **[I3]**
R	R2 R7 R8 R11	• Integrity and authenticity of the data **[R1]** • Documentation of the completeness of the data and metadata **[R1]** • Links to metadata and to other datasets **[R1]** • Provenance data and related audit trails **[R1.2]** • Maintains licenses covering data access and use and monitors compliance **[R1.1]** • Defined data and metadata: ensure relevance and understandability for users **[R1.3]** • Technical data and metadata quality and assessment of adherence to schema **[R1.3]**

Mokrane, 2018

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Data repositories for FAIR data

Doorn, Dijk & Grootveld, 2017

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Thank you !

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